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Human body produces free radicals when it is exposed to x-rays, cigarette smoke, industrial chemicals and air pollutants. Free radicals can disrupt cell membranes, increase the risk of many forms of cancer, damage the interior lining of blood vessels and lead to a high risk of heart disease and stroke. Body uses vitamins and minerals to counter the effects of free radicals. Fruits contains antioxidants which decrease the effects of free radicals.

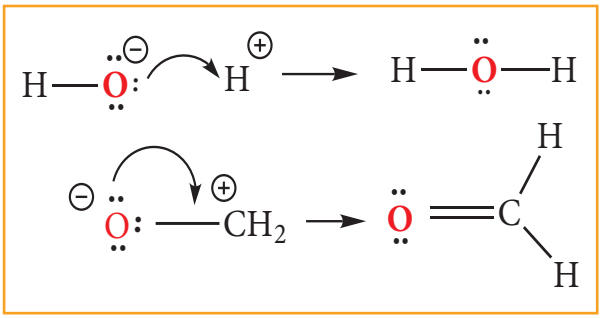
12.1.4 Electron movement in organic reactions

All organic reactions can be understood by following the electron movements, i.e. the electron redistribution during the reaction. The electron movement depends on the nature of the substrate, reagent and the prevailing conditions. The flow of electrons is represented by curved arrows which show how electrons move as shown in the figure. These electron movements result in breaking or formation of a bond (sigma or pi bond). The movement of single electron is indicated by a half -headed curved arrows.

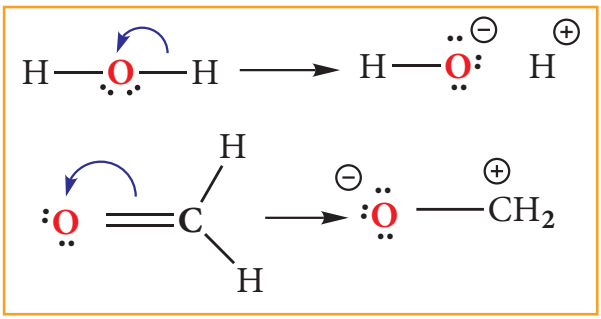
There are three types of electron movement viz.,

- lone pair becomes a bonding pair.
- bonding pair becomes a lone pair
- a bond breaks and becomes another bond.

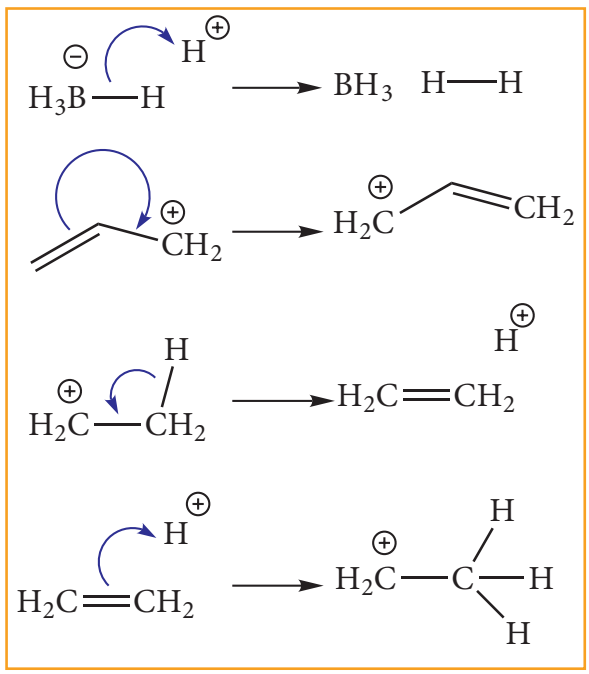
Type 1: A lone pair to a bonding pair



Type 2: A bonding pair to a lone pair



Type 3: A bonding pair to an another bonding pair



12.1.5 Electron displacement effects in co-valent bonds

Some of the properties of organic molecules such as stability, reactivity,